

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem
Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2751–2794; original printed pp. 92–93; physical source-PDF pp. 81–82; recomposed running pp. 73–74. The printed-page-93 marker occurs inside line 2782, after “Whence” and before the concluding observation. Line 2795 is blank; line 2796 begins the second step and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

It follows from formula (2.3) that conditions (a) and (c) are respectively equivalent to:

- (a') for every $q \in \mathbb{Z}$ and every $x \in S'_q$, one has $q + i \neq d(x)$;
(c') for every $q \in \mathbb{Z}$ and every $x \in S'_q$, if $c(x) = 1$, then $q + i \neq d(x)$.

Here is condition (d), promised above:

- (d) for every $q \in \mathbb{Z}$ and every $y \in Z_q$, one has $q + i \neq \dim \mathcal{O}_{X,y}$.

These conditions are equivalent.

(a') $\Rightarrow$ (c') is recorded for completeness.

(d) $\Rightarrow$ (a'). Indeed, let $q \in \mathbb{Z}$ and $x \in S'_q$. Choose a point $y \in \overline{\{x\}} \cap Y$ that realizes the codimension,⁽⁶⁾ that is, such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}) = c(x). \quad (2.5)$$

Since X is regular at y , it follows that

$$\dim \mathcal{O}_{X,y} = d(x) \quad (\text{cf. (2.2)}). \quad (2.6)$$

But $y \in \overline{\{x\}}$, and hence $y \in Z_q$; the conclusion follows.

(c') $\Rightarrow$ (d). Let $q \in \mathbb{Z}$ and $y \in Z_q$. Assume provisionally that there exists $x \in S'_q$ such that

$$y \in \overline{\{x\}} \quad \text{and} \quad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

(One also says that y is an immediate specialization of x .) It follows that $c(x) = 1$, since y realizes the codimension of $\overline{\{x\}} \cap Y$ in $\overline{\{x\}}$, while $x \notin Y$. Condition (c') therefore gives

$$q + i \neq d(x).$$

Whence the conclusion, upon observing that $d(x) = \dim \mathcal{O}_{X,y}$ by (2.6). The assertion provisionally assumed above is the content of the following lemma.

Lemma 2.5. Let X be a locally Noetherian prescheme and Y a closed subset of X . Set $U = X - Y$, and suppose that U is dense in X . For every $y \in Y$, there exists $x \in U$ such that y is an immediate specialization of x , that is,

$$y \in \overline{\{x\}} \quad \text{and} \quad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

We have applied the lemma with X equal to the prescheme $\overline{S'_q}$ and Y equal to the subset $Y \cap \overline{S'_q}$.

Proof of Lemma 2.5. There exists $x \in U$ such that $y \in \overline{\{x\}}$. Choose such an x for which $r = \dim \mathcal{O}_{\overline{\{x\}},y}$ is minimal. We must prove that $r = 1$. By the choice of x , every $z \in \text{Spec}(\mathcal{O}_{\overline{\{x\}},y})$ with $z \neq x$ lies in Y . Consequently, $\{x\}$ is open in $\text{Spec}(\mathcal{O}_{\overline{\{x\}},y})$, and the conclusion follows. $\square$

⁽⁶⁾*Editor's note.* Throughout what follows, closures of points are endowed with the reduced structure.